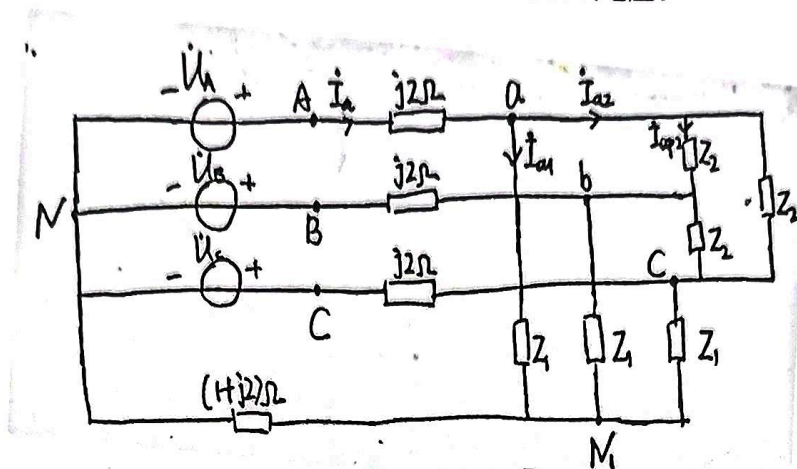


5-1. 如图 5-1 所示电路, 正序对称电源 $\dot{U}_{AB} = 380\angle 0^\circ \text{V}$, $Z_1 = 20\Omega$, $Z_2 = 60\Omega$ 。计算:

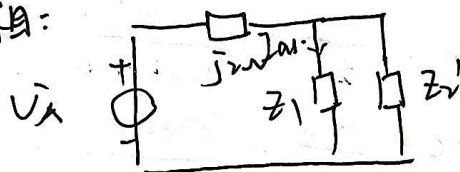
(1) 电流 $\dot{I}_A$ 、 $\dot{I}_{A1}$ 、 $\dot{I}_{A2}$ 、 $\dot{I}_{Ap2}$; (2) 计算负载 Z_1 和 Z_2 的线电压。



(1) 将 Δ 负载 Z_2 等效成 Y 型 $Z_Y' = \frac{Z_Y}{3} = 20\Omega$

$$\dot{U}_{AN} = \frac{1}{\sqrt{3}} \dot{U}_{AB} \angle -30^\circ = 220\angle -30^\circ \text{V}$$

A 相:



$$\dot{I}_A = \frac{220\angle -30^\circ}{j20 + 10} = 21.57\angle -41.31^\circ$$

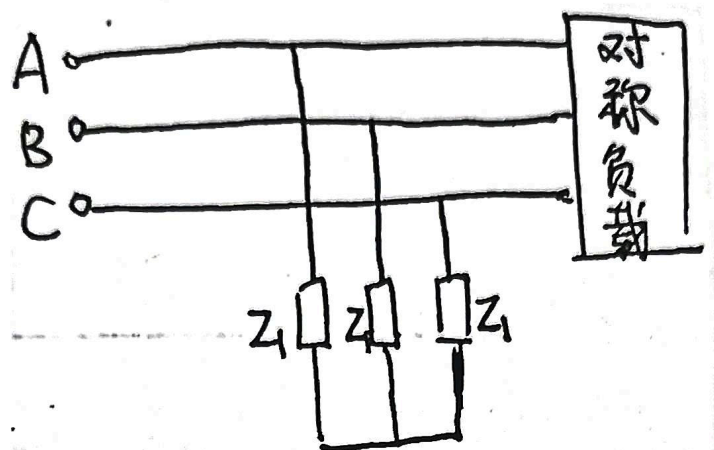
$$\dot{I}_{A1} = \dot{I}_{A2} = \frac{1}{2} \dot{I}_A = 10.78\angle -41.31^\circ$$

$$\dot{I}_{Ap2} = \frac{1}{\sqrt{3}} \dot{I}_A \angle 30^\circ = 6.23\angle -11.31^\circ$$

(2) 相电压 $\dot{U}_{A1} = Z_1 \dot{I}_{A1} = 215.8\angle -41.31^\circ \text{V}$

线电压: $U_{ab} = \sqrt{3} U_{A1} = \sqrt{3} \times 215.8 = 372.6 \text{V}$

5-2. 如图 5-2 所示电路, 工频正序对称电源线电压为 380V。三相负载 $Z_1 = (90 + j120) \Omega$, 另一组对称感性负载的功率因数是 0.6、无功功率是 800var。(1) 计算电源提供的复功率、负载的总功率因数; (2) 计算电源的线电流。



$$11) \lambda = \cos \varphi = 0.6 \quad \varphi > 0 \quad \sin \varphi = 0.8$$

$$\text{对对称负载 } Q = 800 \text{ var} \quad P = 600 \text{ W}$$

$$\tilde{S} = (600 + j800) \text{ V} \cdot \text{A}$$

$$\text{对 } Z_1: U_{AN} = 220 \text{ V} \quad Z_1 = 90 + j120 = 150 \angle 53.1^\circ$$

$$I_A = \frac{220}{150} = \frac{22}{15}$$

$$P_1 = 3 \cdot 220 \times \frac{22}{15} \times \cos 53.1^\circ = 580.8 \text{ W}$$

$$Q_1 = 3 \times 220 \times \frac{22}{15} \times \sin 53.1^\circ = 774.4 \text{ var}$$

$$\tilde{S}_1 = (580.8 + j774.4) \text{ V} \cdot \text{A}$$

$$\tilde{S}_{\Sigma} = (1180.8 + j1574.4) \text{ V} \cdot \text{A} \quad \varphi = \arctan \frac{1574.4}{1180.8} = 53.1^\circ$$

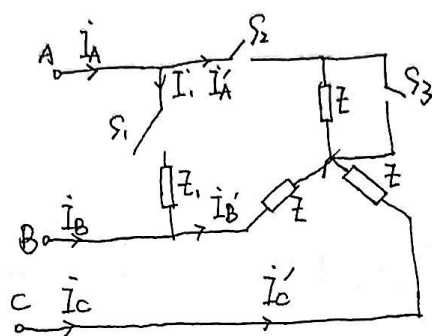
$$\lambda = \cos(53.1^\circ) = 0.6$$

$$12) S = \sqrt{3} U_L I_L \quad \therefore I_L = \frac{\sqrt{(1180.8)^2 + (1574.4)^2}}{\sqrt{3} \times 380} = 2.98 \text{ A}$$

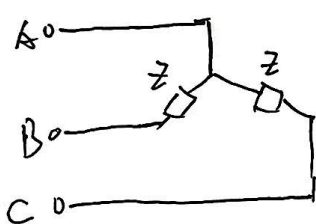
5-3 如图 5-3 所示，图中对称三相电源的线电压为 380V，负载

$Z = 100 + j100\Omega$, $Z_1 = 50 + j50\Omega$ ，求

- (1) 开关 S_1 打开， S_2 、 S_3 合上时的线电流
- (2) 开关 S_1 、 S_2 、 S_3 都打开时的线电流
- (3) 开关 S_3 打开， S_1 、 S_2 合上时的线电流



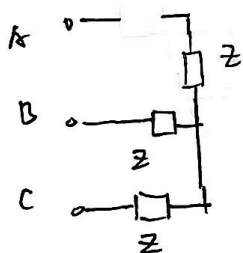
1) S_1 打开， S_2 、 S_3 闭合



B、C 线电流 $I_B = I_C = \frac{380}{\sqrt{100^2 + 100^2}} = 2.68 \text{ A}$

A 线电流 $I_A = \sqrt{3} I_B = 4.65 \text{ A}$

2) S_1 、 S_2 、 S_3 打开

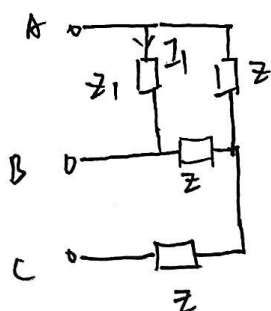


A 相断路 $I_A = 0$

B、C 相线电流

$I_B = I_C = \frac{380}{2|Z|} = \frac{380}{2 \times 100\sqrt{2}} = 1.34 \text{ A}$

3) S_3 打开， S_1 、 S_2 闭合



设 $U_{AB} = 380 \angle 0^\circ$ $U_{AN} = 220 \angle -30^\circ \text{ V}$

$U_{BN} = 220 \angle -150^\circ \text{ V}$ $U_{CN} = 220 \angle 90^\circ \text{ V}$

$I_1 = \frac{380 \angle 0^\circ}{50 + j50} = 5.37 \angle -45^\circ \text{ A}$

$I_{A'} = \frac{220 \angle -30^\circ}{100 + j100} = 1.55 \angle -75^\circ \text{ A}$

$I_{B'} = 1.55 \angle 165^\circ \text{ A}$ $I_{C'} = 1.55 \angle 45^\circ \text{ A}$

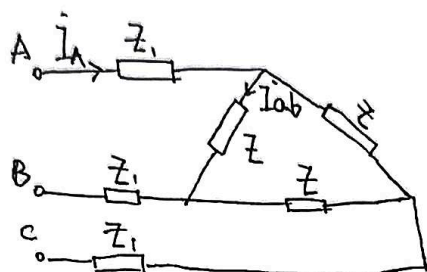
$I_A = I_{A'} + I_1 = 6.76 \angle -51.59^\circ \text{ A}$ $I_B = I_{B'} - I_1 = 6.76 \angle 41.59^\circ \text{ A}$

$I_C = 1.55 \angle 45^\circ \text{ A}$

5-4 如图 5-4 所示, 试问下列算式哪些对哪些不对

$$(1) i_{ab} = \frac{U_{AB}}{Z} \quad (2) i_{ab} = \frac{U_{AB}}{Z + Z_L} \quad (3) i_{ab} = \frac{U_{AB}}{2Z_L + Z}$$

$$(4) I_A = \frac{U_{AB}}{Z_L + Z/3} \quad (5) I_A = \frac{\dot{U}_{AN}}{Z_L + Z/3} \quad (6) I_A = \frac{U_{AB}}{2Z_L + Z}$$



11) × 12) × 13) × 14) × 15) ✓ 16) ×

$$I_{ab} = \frac{U_{AB}}{Z + 3Z_L}$$

41.3°